

P520 Proxmox VE — Complete Build Runbook

Verified working command sequence · OSIM v2.0 · 2026-03-28 nvme1n1 (OS/ZFS) + nvme0n1 (XFS/Wazuh) + sda+sdb (ZFS Mirror/PBS)

***DISCLAIMER:** All IP addresses, hostnames, passwords, and credentials shown in this document have been replaced with generic placeholders for public distribution. Values shown in angle brackets (e.g. <PROXMOX-IP>, <YOUR-STRONG-PASSWORD>) are NOT real — substitute your own values before use. Never use placeholder passwords in a production environment.*

Component	Spec	Role
nvme1n1	2TB Samsung 990 EVO Plus	Proxmox OS + VM disks (ZFS, 200GB OS partition)
nvme0n1	1TB TEAM	Wazuh/OpenSearch data (XFS, full disk)
sda	465GB Seagate	PBS backup mirror — ZFS mirror with sdb
sdb	465GB WD	PBS backup mirror — ZFS mirror with sda

Step 1 — Flash Proxmox USB Installer (MacBook)

Download Proxmox VE 8.3 ISO from proxmox.com/downloads. Flash to USB using dd.

```
diskutil list
diskutil unmountDisk /dev/disk4
sudo dd if=/Downloads/proxmox-ve_8.3-1.iso of=/dev/rdisk4 bs=1m status=progress
diskutil eject /dev/disk4
```

Step 2 — BIOS Configuration

Boot P520, press F1 for BIOS. Press F12 to select USB boot device.

```
Boot Mode:      UEFI only
Secure Boot:    Disabled
Hyperthreading: Enabled
VT-d:          Enabled
```

Step 3 — Proxmox Installer — Disk Selection

Select nvme1n1 (2TB Samsung) ONLY. Click Options before confirming disk.

```
Filesystem:    ZFS (RAID0)
Target Disk:   nvme1n1 (2TB Samsung 990 EVO — ONLY this disk)
hdspace:       200
ashift:        12
compress:      lz4
checksum:      sha256
copies:        1
```

Step 4 — Proxmox Installer — Network and Credentials

```
Hostname:    p520.internal
IP Address:  <PROXMOX-IP>
Netmask:     255.255.255.0
Gateway:     <GATEWAY-IP>
DNS:         <GATEWAY-IP>
```

Step 5 — Fix: Old ZFS Pool Conflict (initramfs)

If installer drops to initramfs emergency shell due to existing rpool on other disks:

```
dd if=/dev/zero of=/dev/sda bs=1M count=100
dd if=/dev/zero of=/dev/sdb bs=1M count=100
dd if=/dev/zero of=/dev/nvme0n1 bs=1M count=100
reboot -f
# Then retry Proxmox installer from USB
```

Step 6 — Post-Install: Disable Enterprise Repos

```
sed -i 's/^deb/# deb/' /etc/apt/sources.list.d/pve-enterprise.list
sed -i 's/^deb/# deb/' /etc/apt/sources.list.d/ceph.list
rm /etc/apt/sources.list.d/pve-enterprise.sources
rm /etc/apt/sources.list.d/ceph.sources
echo "deb http://download.proxmox.com/debian/pve bookworm pve-no-subscription" \
  > /etc/apt/sources.list.d/pve-no-sub.list
apt update && apt dist-upgrade -y
```

Step 7 — Storage Setup — XFS on nvme0n1 (Wazuh/OpenSearch)

```
fdisk /dev/nvme0n1 << EOF
g
n
1

w
EOF

mkfs.xfs /dev/nvme0n1p1
mkdir -p /mnt/wazuh-data
echo "/dev/nvme0n1p1 /mnt/wazuh-data xfs defaults,noatime,nodiratime 0 2" \
  >> /etc/fstab
mount /mnt/wazuh-data
systemctl daemon-reload
df -h /mnt/wazuh-data # should show ~954GB
```

Step 8 — Storage Setup — ZFS Mirror on sda+sdb (PBS Backups)

```
dd if=/dev/zero of=/dev/sda bs=1M count=100
dd if=/dev/zero of=/dev/sdb bs=1M count=100
```

```
zpool create -f \
-o ashift=12 \
-O compression=lz4 \
-O checksum=sha256 \
-O atime=off \
pbspool mirror /dev/sda /dev/sdb
```

```
zpool status pbspool    # verify ONLINE
```

Step 9 — Register Storage Pools in Proxmox

```
pvesm add dir wazuh-data --path /mnt/wazuh-data --content images,rootdir
pvesm add zfspool pbspool --pool pbspool --content images,rootdir
zfs create rpool/vmdata
pvesm add zfspool vmdata --pool rpool/vmdata --content images,rootdir
```

```
# Tune ZFS ARC to 16GB max
echo "options zfs zfs_arc_max=17179869184" > /etc/modprobe.d/zfs.conf
echo 17179869184 > /sys/module/zfs/parameters/zfs_arc_max
```

```
pvesm status    # verify all 5 pools active
```

Step 10 — Organize Proxmox GUI — Pools and Tags

```
pvesh create /pools --poolid SECURITY --comment "VLAN 40 - Wazuh, Detection Stack"
pvesh create /pools --poolid OPS --comment "VLAN 20 - Wiki, Gitea, MISP, TheHive"
pvesh create /pools --poolid MGMT --comment "VLAN 11 - PBS, Infrastructure"
pvesh create /pools --poolid LAB --comment "VLAN 20 - Blue Team Targets"
pvesh create /pools --poolid RED --comment "VLAN 50/51/52 - Red Node"

pvesh set /cluster/options --tag-style \
'color-map=security:ef4444;ops:3b82f6;mgmt:8b5cf6;lab:f59e0b;red:dc2626;\
wazuh:10b981;opensearch:06b6d4;misp:f97316;backup:6b7280;detection:1e3a5f'

pvesh set /cluster/options --registered-tags \
"security;ops;mgmt;lab;red;wazuh;opensearch;misp;backup;detection;zeek;suricata;caldera;kali;pbs"

pvesh set /cluster/options --description \
"OSIM v2.0 — Zero Trust Security Lab | Five-Zone Architecture | Default-Deny Enforcement"
```

Step 11 — Install Tailscale on P520

```
curl -fsSL https://tailscale.com/install.sh | sh
tailscale up --accept-dns=false
# Authorize node at the URL provided
tailscale status    # verify p520 shows as connected
```

Step 12 — Enable VLAN-Aware Bridge

Edit /etc/network/interfaces — add bridge-vlan-aware yes to vmbr0 section:

```
auto vmbr0
iface vmbr0 inet static
    address <PROXMOX-IP>/24
    gateway <GATEWAY-IP>
    bridge-ports nic0
    bridge-stp off
    bridge-fd 0
    bridge-vlan-aware yes

ifreload -a
```

Step 13 — Deploy Wazuh VM via Cloud-Init

Note: Use virtio-scsi-pci controller — lsi controller causes initramfs boot failure with cloud images.

```
# Download Ubuntu 22.04 cloud image
cd /var/lib/vz/template/iso/
wget -O jammy-server-cloudimg-amd64.img \
    https://cloud-images.ubuntu.com/jammy/current/jammy-server-cloudimg-amd64.img

# Create VM
qm create 100 --name wazuh-manager --memory 8192 --balloon 0 --cores 4 \
    --cpu host --net0 virtio,bridge=vmbr0 --ostype l26 \
    --tags "wazuh;security;detection" --pool SECURITY \
    --serial0 socket --vga serial0 \
    --description "Wazuh Manager | VLAN 40 SECURITY | <SECURITY-IP>"

# Import cloud image
qm importdisk 100 /var/lib/vz/template/iso/jammy-server-cloudimg-amd64.img local-zfs

# CRITICAL: use virtio-scsi-pci not lsi
qm set 100 --scsihw virtio-scsi-pci
qm set 100 --scsi0 local-zfs:vm-100-disk-0,discard=on,ssd=1
qm set 100 --ide2 local-zfs:cloudinit
qm set 100 --boot order=scsi0
qm resize 100 scsi0 32G

# Configure cloud-init
qm set 100 --ciuser osim
qm set 100 --cipassword $(openssl passwd -6 '<YOUR-STRONG-PASSWORD>')
qm set 100 --ipconfig0 ip=<WAZUH-VM-IP>/24,gw=<GATEWAY-IP>
qm set 100 --nameserver <GATEWAY-IP>
qm set 100 --searchdomain internal

# Start VM
qm start 100
```

Step 14 — Fix VM Network — Permanent Netplan Config

From VM console (qm terminal 100 on P520). Log in as osim.

```
sudo nano /etc/netplan/50-cloud-init.yaml
```

```
# File contents:
network:
  version: 2
  ethernet:
    ens18:
      addresses:
        - <WAZUH-VM-IP>/24
      routes:
        - to: default
          via: <GATEWAY-IP>
      nameservers:
        addresses:
          - <GATEWAY-IP>
        search:
          - internal
      dhcp4: false
```

```
sudo netplan apply
```

```
sudo reboot
```

Step 15 — Enable SSH Password Auth and Add sysadmin User

Note: Cloud-init disables password auth via /etc/ssh/sshd_config.d/60-cloudimg-settings.conf override.

```
# Fix SSH password auth
sudo sed -i 's/PasswordAuthentication no/PasswordAuthentication yes/' \
  /etc/ssh/sshd_config.d/60-cloudimg-settings.conf
sudo systemctl restart ssh
```

```
# Add sysadmin user
sudo adduser sysadmin
sudo usermod -aG sudo sysadmin
```

Step 16 — Verify SSH Access from MacBook

```
# Remove stale host key if needed
ssh-keygen -R <WAZUH-VM-IP>

# SSH in as sysadmin
ssh sysadmin@<WAZUH-VM-IP>

# Optional: copy SSH key for passwordless access
ssh-copy-id sysadmin@<WAZUH-VM-IP>
```

Build complete. Next steps: Wazuh all-in-one install on VM 100 · VLAN 40 migration · Zeek · MISP · TheHiveVLAN 40 migration · Zeek · MISP · TheHive